

Propagation-gate §1.1 paired meta-tests — evidence

Date: 2026-06-23 **Scope:** Bluff-proof the CM-COVENANT-114-N-PROPAGATION anchor-presence gate class in tests/pre_build_verification.sh (audit found them real-but-UNPAIRED). **New file:** tests/meta/meta_test_propagation_gates.sh (sources lib_metatest.sh).

What the propagation gates are

Each gate (pre_build_verification.sh lines 37-50) is the literal command `grep -qF '11.4.N' CLAUDE.md` — PASS iff the anchor literal is present in the project context carrier, FAIL iff absent.

Sample paired (across the range)

Gate	Anchor	clean→PASS	strip-anchor→FAIL	restore→PASS
CM-COVENANT-114-153-PROPAGATION	11.4.153 (early)	rc=0	rc=1	rc=0
CM-COVENANT-114-159-PROPAGATION	11.4.159 (middle)	rc=0	rc=1	rc=0
CM-COVENANT-114-166-PROPAGATION	11.4.166 (newest)	rc=0	rc=1	rc=0

Mutation = `sed 's/11\.4\.N/STRIPPED_FOR_META/g'` over the REAL CLAUDE.md (strips ALL occurrences; CLAUDE.md repeats each anchor). The EXACT gate command (`grep -qF '11.4.N' CLAUDE.md`) is invoked as subject — no inline grep replica.

Result

- `meta_test_propagation_gates.sh` exit 0 — every case mutate→FAIL→restore→PASS.
- `tests/meta/run_all.sh` exit 0 — **5/5 gates proven bluff-proof** (was 4 from F-METAGATES, +1 propagation-gate meta-test).
- `bash -n / sh -n clean (§11.4.67)`.

- CLAUDE.md restored byte-identical: work blob sha256
e20e8c4f90488f7dce68033157b4178a6915a6f6 == HEAD:CLAUDE.md blob
— no mutation residue (§11.4.84).

Honest boundary (§11.4.6)

The pre-build propagation gates read CLAUDE.md only; this meta-test mutates CLAUDE.md — the exact file the gate reads. AGENTS.md/GEMINI.md §11.4.157
lockstep is a separate, not-yet-pre-build-wired concern and is NOT claimed here.

Logs

- meta_test_propagation_gates.run.log
- run_all.run.log